

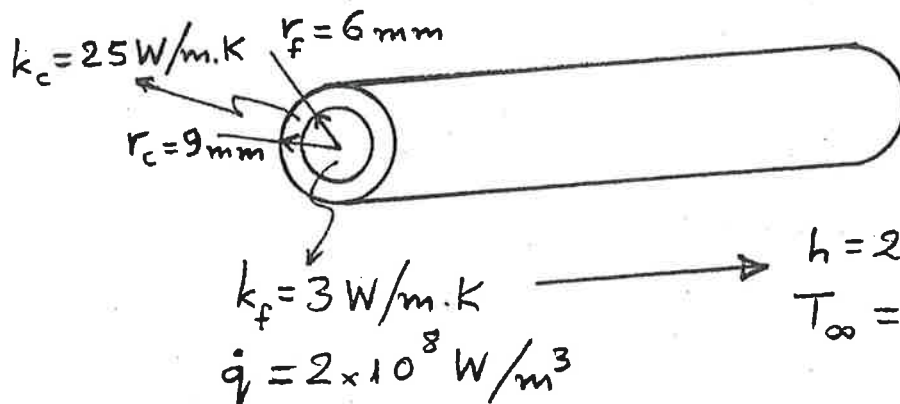
COURSE: CHE 312: Transport Phenomena II**Spring Semester 2020****Department of Chemical Engineering****University of Illinois at Chicago****Chicago, Illinois 60607-7052****Quiz #2****February 12, 2020*****Closed Notes and Book*****Duration: 25 minutes**

3/10

 r
 $-r$ $h(T - T_\infty)$

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
- Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?


 $\dot{q} = \text{constant}$
 $k_f = \text{constant}$
 $k_c = \text{constant}$

$$T = -\frac{q r^2}{4k} + \left(q'' r_f - \frac{q r_f^2}{2} \right) \ln r/k + T_S$$

$$= -\left[\frac{2 \cdot 10^8 \text{ W/m}^2 (0.006^2)}{4 (3 \text{ W/m} \cdot \text{K})} \right] + \left(q'' (0.006 \text{ m}) - \frac{2 \cdot 10^8 \text{ W/m}^2 (0.006^2)}{2} \right) \frac{\ln 0.006}{3} + T_S$$